Example 8.24

Find the area of the parallelogram whose adjacent sides are $\vec{a} = 3\hat{i} + \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.

Solution

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 4 \\ 1 & -1 & 1 \end{vmatrix} = \hat{i}(1+4) - \hat{j}(3-4) + \hat{k}(-3-1) = 5\hat{i} + \hat{j} - 4\hat{k} \ .$$

$$|\vec{a} \times \vec{b}| = |5\hat{i} + \hat{j} - 4\hat{k}| = \sqrt{42}$$

Area of the parallelogram is $\sqrt{42}$ sq.units.

Example 8.25

For any two vectors $\vec{a}$ and $\vec{b}$, prove that $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 \ |\vec{b}|^2$

Solution

We have, $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}| \sin\theta$ and $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$

$$|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 \ |\vec{b}|^2 \sin^2\theta + |\vec{a}|^2 \ |\vec{b}|^2 \cos^2\theta$$

$$= |\vec{a}|^2 \ |\vec{b}|^2 (\sin^2\theta + \cos^2\theta) = |\vec{a}|^2 \ |\vec{b}|^2 .$$

Example 8.26

Find the area of a triangle having the points $A(1,0,0), B(0,1,0),$ and $C(0,0,1)$ as its vertices.

Solution :

Let us find two sides of the triangle.

$$\overrightarrow{AB} = -\hat{i} + \hat{j} \ ; \ \overrightarrow{AC} = -\hat{i} + \hat{k}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}$$

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{3}$$

The area of the triangle ABC is $\frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{\sqrt{3}}{2}$.

Note 8.8

Instead of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, one can take any two sides.

EXERCISE 8.4

(1) Find the magnitude of $\vec{a} \times \vec{b}$ if $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} + 5\hat{j} - 2\hat{k}$.

(2) Show that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = \vec{0}$.

(3) Find the vectors of magnitude $10\sqrt{3}$ that are perpendicular to the plane which contains $\hat{i} + 2\hat{j} + \hat{k}$ and $\hat{i} + 3\hat{j} + 4\hat{k}$.

(4) Find the unit vectors perpendicular to each of the vectors $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.

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